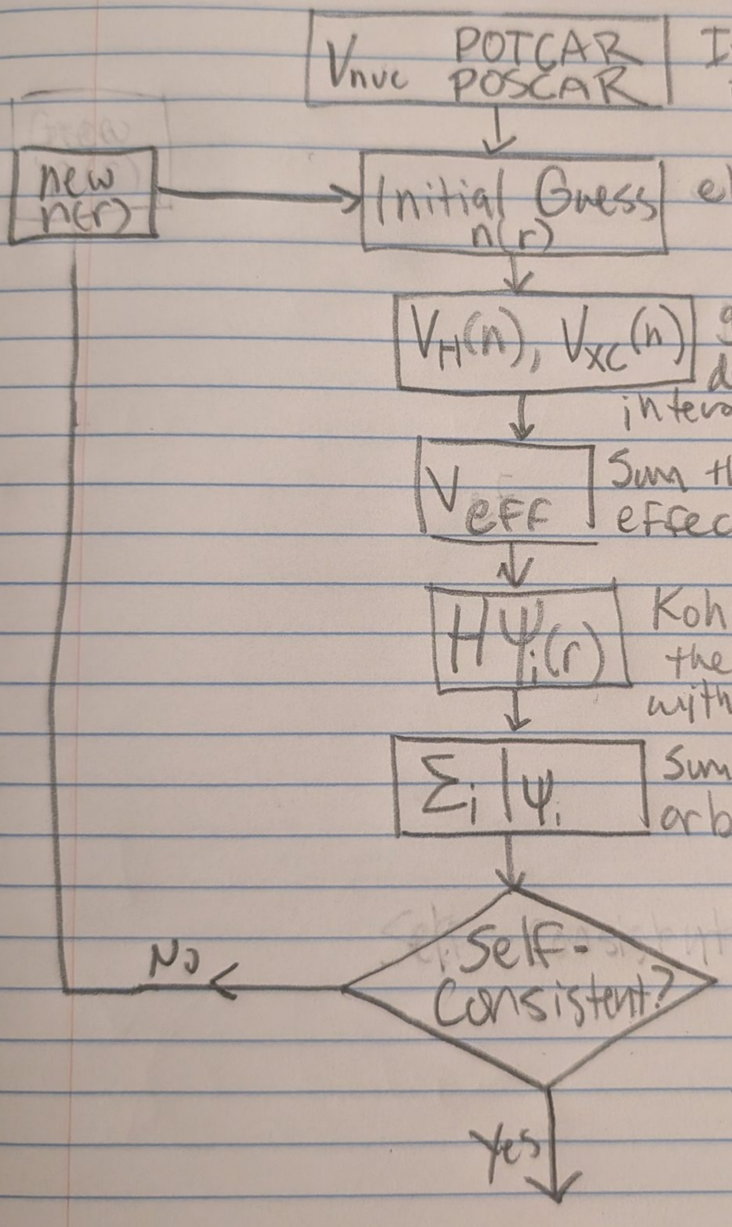
Homework 2

Dallin Gordon

Section 1 (Questions 1–3): handwritten responses, pages 1–5. Section 2: Problem 4 and Problem 5 (both completed). Code is submitted separately in the zip file.


V_{nuc} POTCAR POSCAR
new $n(r)$
Initial Guess $n(r)$
$V_H(n)$, $V_{xc}(n)$
V_{eff}
$H\Psi_i(r)$
$\Sigma_i |\Psi_i$
No
Self-Consistent?
Yes
outterloop step

Initial potentials, positions,

electron density guess

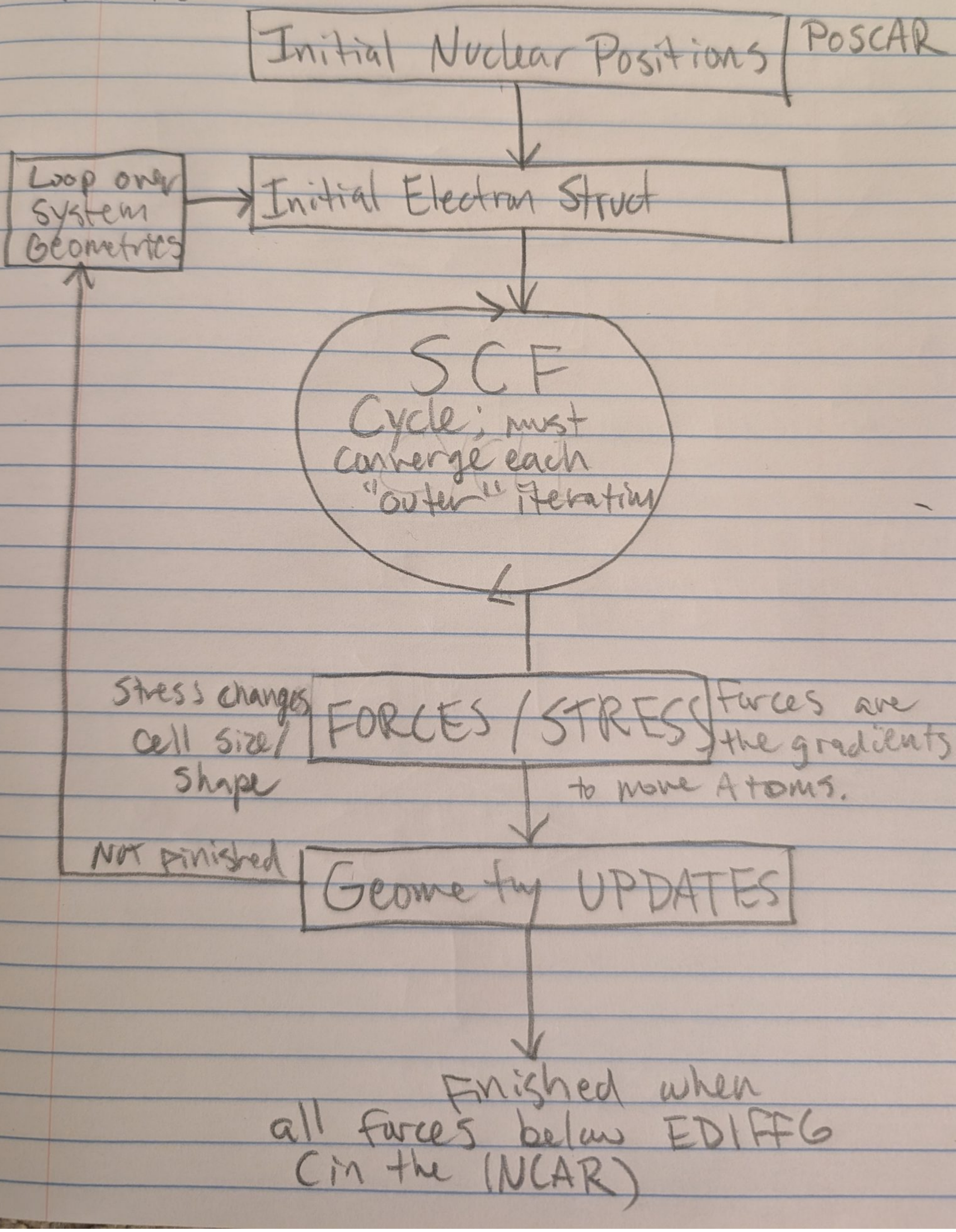
given the electron densities, how they interact (Hartree, quantum)

Sum then with proton effects

Koh-sham, the Hamiltonian the wave function, but with our bases!

Sum over occupied orbitals with our Ψ

Q, cont'd


Initial Nuclear Positions
POSCAR
Loop over system Geometries
Initial Electron Struct
SCF
Cycle; must converge each "outer" iteration
Stress changes cell size/shape
FORCES / STRESS
Forces are the gradients to move Atoms.
Not finished
Geometry UPDATES
Finished when all forces below EDIFFG (in the INCAR)

Q2

Force is the negative gradient of energy

$$F_I = -\frac{\partial E_{tot}}{\partial R_1}$$

seperating that further we have:

$$F_I = F_I^{HF} + F_I^{P} + F^{other}$$

where HF is Hellman Feynman;

$$F_I^{HF} = -\int n(r) \frac{\partial V_{ext}(r, \{R_I\})}{\partial R_I} dr$$

The forces on the nuclei by converged electronic charge density. P is Pulay;

$$F_I^{P} = -\sum_{\mu} \frac{\partial \phi_{\upsilon}}{\partial R_I} \frac{\partial E_{tot}}{\partial \phi_{\upsilon}}$$

and This is where our choise of basis is involved. This is chain rule, the second term is how energy changes as our basis functions change, The first term is how our bases are affected by nuclear motion. (Gaussian bases (and other atom centered bases) change as the atoms move, F_I^P decays or vanishes when the bases dont change with atom movement. (plane waves mentiond in the slides)

Q3

meta GGA is GGA with kinetic energy $\tau(r)$ included;

$$E_{XC}^{MGGA}(n) = \int n(r)\,\varepsilon_{xc}(n(r), \nabla n(r), \tau(r))\,dr$$

with $\tau(r) = \frac{1}{2}\sum^{occ} |\nabla \phi_i(r)|^2$, the gradient of our bases function.

This is slightly more compute heavy than GGA. τ is good for recognizing metallic or coraleat regions.

Hybrid:

$$E_x^{hyb}(n) = \alpha E_x^{HF} + (1-\alpha) E_x^{GGA}(n) + E_c^{GGA}(n)$$

which adds in exact (Hartree-Frock) orbitals, only estimated at "lower rungs". This layer gets more accurate band gaps, suited to semi-conductors and insulators. Good for ceramics and polymers, lower levels "smear" the density function to such a degree the features of these materials are lost in the estimate. The E^{HF} term;

$$E_x^{HF} = -\frac{1}{2}\sum_{ij}^{occ} \iint \frac{\psi_i^*(r)\,\psi_j^*(r')\,\psi_j(r)\,\psi_i(r')}{|r-r'|}\,dr\,dr'$$

with its double integration and pairwise electron operations is not trivial, compute demands grow with the input. (it appears at least gradratic growth to me)

Q3 contd

RPA;

$$E_{xc}^{RPA} = E_x^{EXX} + E_c^{RPA}$$

E_x^{EXX} is exact exchange, with $\alpha = 1.0$ from hybrid, and now the RPA correlation;

$$E_c^{RPA} = \frac{1}{2\pi}\int_0^{\infty} d\omega\, \mathrm{Tr}\left(\ln(1-\chi_0(i\omega)v) + \chi_0(i\omega)v\right)$$

where χ_0 is the response function; and

$$\chi_0(r, r', i\omega) = \sum_i^{occ}\sum_a^{unocc} \frac{\psi_i^*(r)\psi_a(r)\psi_a^*(r')\psi_i(r')}{i\omega - (\varepsilon_a - \varepsilon_i)} + cc$$

So this allows us to capture perturbations through the Coulomb interaction. This does well for 2d surfaces, layered materials. now you add additional sums over pairs of occupied and unoccupied orbitals. this grows faster than quadratic.

Problem 4 — Geometry optimization of distorted Si

INCAR

```
SYSTEM = Distorted Si geometry optimization
ENCUT = 400
EDIFF = 1E-6
EDIFFG = -0.02
IBRION = 2
ISIF = 2
NSW = 30
PREC = Accurate
LREAL = Auto
```

KPOINTS

```
Gamma-centered 5x5x5 mesh for distorted Si
0
Gamma
5 5 5
0 0 0
```

Summary table

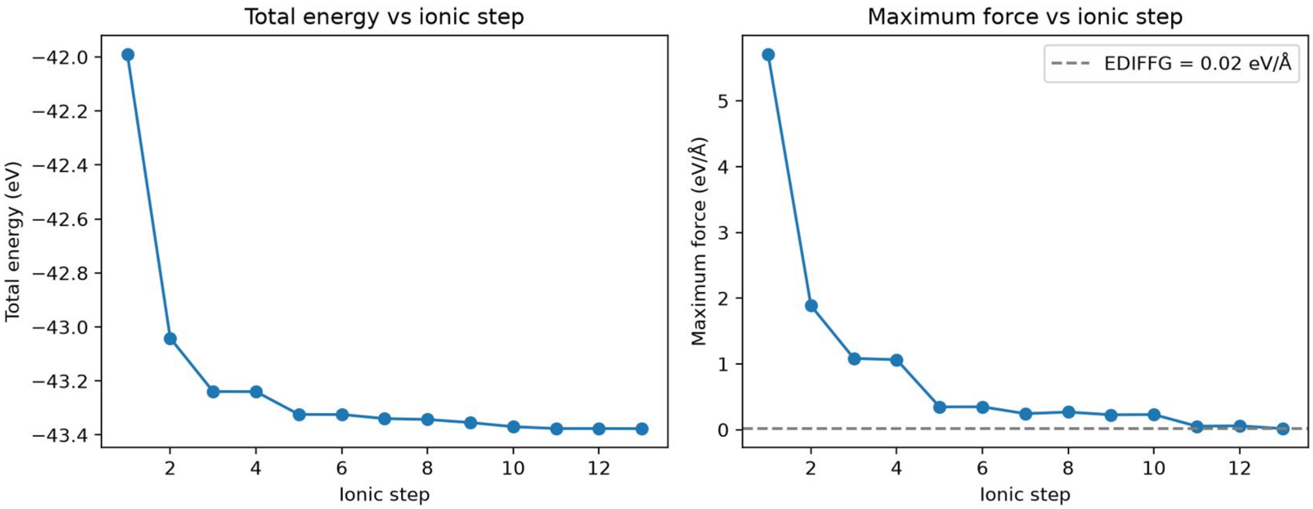
Quantity	Value
Number of ionic steps	13
Initial energy (eV)	-41.9898
Final energy (eV)	-43.3768
Final maximum force (eV/Å)	0.0157
Largest atomic displacement (Å)	0.1882
Atom with largest displacement	5
SCF iterations, first ionic step	13
SCF iterations, last ionic step	4

Script output

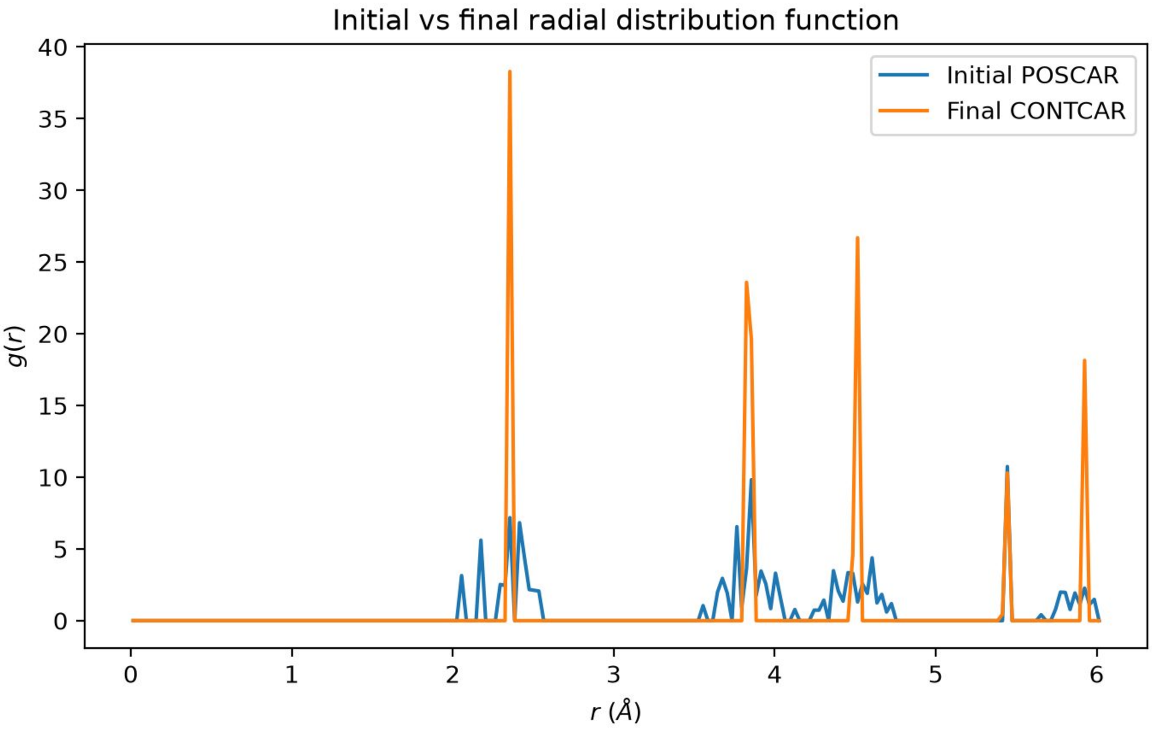
```
Per-atom displacement (A):
  atom 1 (Si): 0.0249    atom 5 (Si): 0.1882
  atom 2 (Si): 0.1009    atom 6 (Si): 0.1560
  atom 3 (Si): 0.1254    atom 7 (Si): 0.1752
  atom 4 (Si): 0.1123    atom 8 (Si): 0.1801
SCF iterations per ionic step: [13, 9, 7, 5, 7, 4, 6, 6, 5, 6, 6, 3, 4]
```

Convergence: OUTCAR reports *reached required accuracy - stopping structural energy minimisation* after 13 of NSW = 30 ionic steps.

Energy and maximum force vs. ionic step



Radial distribution function, initial vs. final



Interpretation

1. Why must the electronic structure be converged before VASP can update atomic positions?

If the electronic structure (the inner loop) hasn't converged, the gradients used to update atomic positions would be incorrect, and future iterations would lead to compounding errors. It seems like Expectation Maximization to me, but one of the EM steps is fully converging.

2. Why does the total energy generally decrease during the optimization?

This is a result of our modelling, particles want to go to lower energy states. So we set our optimizer to minimize energy. It will tend toward LOCAL minima based on the gradient of the loss function (energy). Sufficiently large gradient steps can over shoot and end up in higher energy configurations, so the loss can be noisy and not strictly decrease during relaxation.

3. Why is a force criterion (EDIFFG) more useful than stopping when energy changes only slightly?

A small change in energy doesn't mean we have minimized we may be at a plateau. We are more interested in a configuration where the force (the gradient of the loss, energy) on each atom is low, in our case below 0.02 per the EDIFFG parameter.

4. Based on final forces and RDF, is the final structure relaxed to the crystalline ground state?

Our final max atomic force is 0.016, below our EDIFFG value, and so we are at least at a local minima. However our RDF png looks a whole lot like what we expect Silicon to look like, which suggests we are at a useful one.

Problem 5 — DOS of FCC Al vs. diamond Si

Settings used

	Al	Si
Structure	1-atom primitive FCC, a = 4.05 Å	relaxed CONTCAR from Problem 4 (8 atoms)
SCF k-mesh	11×11×11 Gamma	5×5×5 Gamma
DOS k-mesh	21×21×21 Gamma	9×9×9 Gamma
SCF INCAR	ENCUT=400, EDIFF=1E-6, PREC=Accurate, LREAL=Auto, LWAVE/LCHARG=.TRUE.	same
DOS INCAR	+ ICHARG=11, NEDOS=2000, ISMEAR=0, SIGMA=0.05 eV	same

SCF convergence

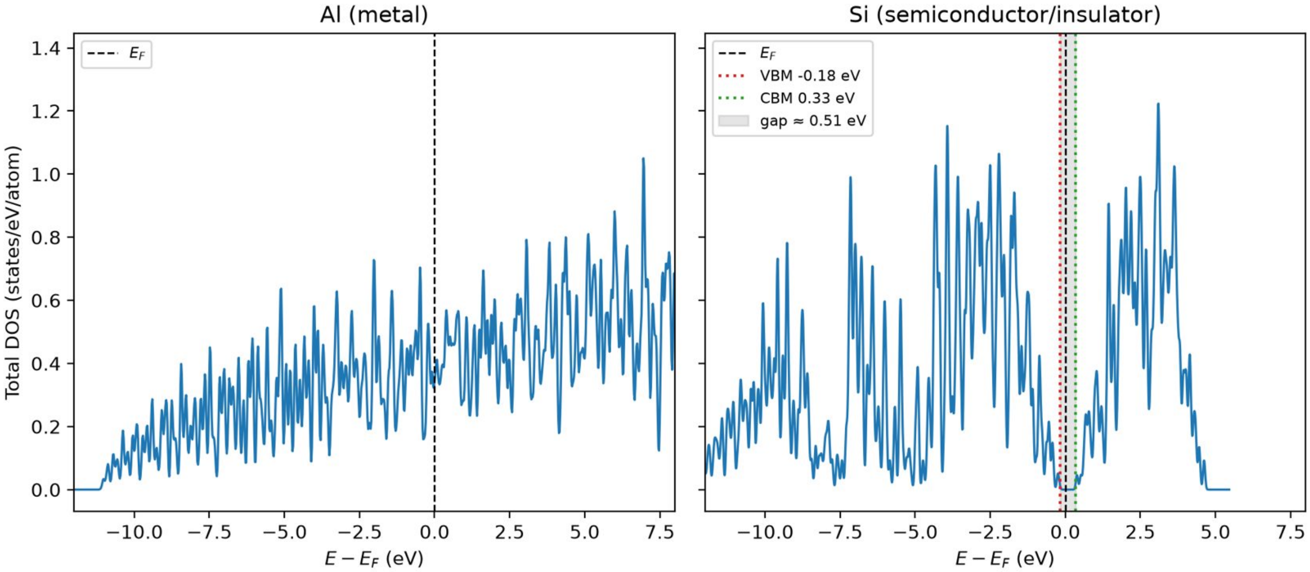
Al: converged in 10 electronic steps, F = –3.754 eV. Si: converged in 11 electronic steps, F = –43.377 eV.

Comparison table

Material	Fermi energy (eV)	DOS at E_F (states/eV/atom)	Calculated gap (eV)	Classification
Al	7.949	0.344	0	metal
Si	6.034	0.000	0.51	semiconductor

Si band edges: VBM = –0.18 eV, CBM = +0.33 eV relative to E_F. Experimental gap ≈ 1.1 eV.

Al vs. Si density of states



Interpretation

1. What feature of the DOS near the Fermi level distinguishes aluminum from silicon?

The DOS at E_F for aluminum has a positive value while there is a gap between positive values around E_F for silicon where the DOS is zero.

2. Why can aluminum conduct electrically without electrons first being excited across an energy gap?

Aluminum has unoccupied states contiguous to E_F requiring very little energy of an electron to enter. The gap has no states. Electrons have nothing to occupy.

3. How does the calculated silicon band gap compare with experiment?

We calculated just about half the experimental gap (0.508 eV). This boils down to the model we chose (PAW_PBE in the POTCAR). DFT is calculating ground states and doesn't take into consideration excited states, which as seen here results in underestimation.

4. Would the DOS alone be sufficient to determine whether an unknown material is metallic or semiconducting? Explain what feature you would examine.

the DOS at E_F; I imagine there would be plenty of false negatives, semiconducting materials with smaller gaps could appear as metallic.